STUDENT UTILITY SYSTEM

Java Mini Project Report

1. Aim

To develop a menu-driven Java application called Student Utility System that performs student-details processing and common mathematical operations. The project demonstrates classes, objects, methods, keyboard input, conditional statements, switch-case, loops, and arithmetic operations.

2. Description

The application displays a menu containing 14 choices. The user selects an option by entering its number. A switch statement calls the corresponding method in the StudentUtility class. The menu repeats through a do-while loop until the user selects option 14 to exit. The system includes student marks processing, number checks, grade calculation, day-name display, multiplication tables, number series, factorial, prime checking, maximum values, and circle-area calculation.

Functions

1. Student Details
2. Even or Odd
3. Largest of Three Numbers
4. Grade Calculator
5. Day Name
6. Multiplication Table
7. Print Numbers
8. Sum of Natural Numbers
9. Fibonacci Series
10. Factorial
11. Prime Number
12. Maximum of Two Numbers
13. Area of Circle
14. Exit

3. Software Requirements

• Programming language: Java
• JDK: JDK 8 or later recommended
• IDE/editor: IntelliJ IDEA, Eclipse, NetBeans, VS Code, or any Java-capable editor
• Execution environment: Computer with JDK/JRE installed
• Input/output: Keyboard input using Scanner and text output in the console
• Libraries: java.util.Scanner and the built-in Math.PI constant

4. Algorithm

Main algorithm

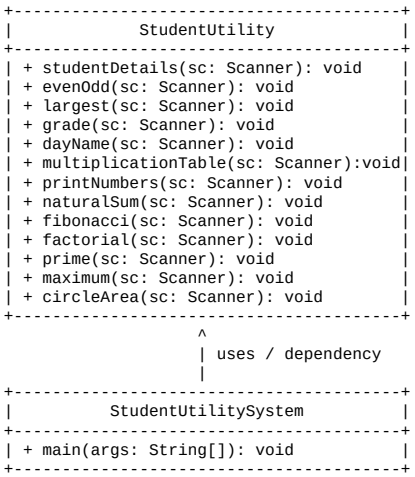
1. Start the program.
2. Create a Scanner object and a StudentUtility object.
3. Display the menu and read the user's choice.
4. Use switch-case to call the method corresponding to the selected choice.
5. If choice 14 is selected, display the exit message.
6. If an invalid choice is entered, display an error message.
7. Repeat the menu until choice 14 is entered.
8. Close the Scanner and stop the program.

Example algorithm: Even or Odd

1. Read an integer n.
2. If n % 2 equals 0, display that n is even.
3. Otherwise, display that n is odd.
4. Return to the main menu.

5. UML Class Diagram

StudentUtilitySystem creates and uses an object of StudentUtility to run the selected utility method.



6. Source Code

Save the source as StudentUtilitySystem.java in the observation folder because it begins with package observation;.

```
package observation;
import java.util.Scanner;

class StudentUtility {
    void studentDetails(Scanner sc) {
        System.out.print("Enter Student Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Roll Number: ");
        int rollNo = sc.nextInt();

        System.out.print("Enter Marks in 3 Subjects: ");
        int m1 = sc.nextInt();
        int m2 = sc.nextInt();
        int m3 = sc.nextInt();

        int total = m1 + m2 + m3;
        double percentage = total / 3.0;

        System.out.println("\n--- Student Details ---");
        System.out.println("Name       : " + name);
        System.out.println("Roll No    : " + rollNo);
        System.out.println("Total Marks: " + total);
        System.out.printf("Percentage : %.2f%%\n", percentage);
    }

    void evenOdd(Scanner sc) {
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        if (n % 2 == 0)
            System.out.println(n + " is Even");
        else
            System.out.println(n + " is Odd");
    }

    void largest(Scanner sc) {
        System.out.print("Enter three numbers: ");
        int a = sc.nextInt(), b = sc.nextInt(), c = sc.nextInt();
        int large = a;
        if (b > large) large = b;
        if (c > large) large = c;
        System.out.println("Largest Number = " + large);
    }

    void grade(Scanner sc) {
        System.out.print("Enter Percentage: ");
        double percentage = sc.nextDouble();
        if (percentage >= 90) System.out.println("Grade: A");
        else if (percentage >= 80) System.out.println("Grade: B");
        else if (percentage >= 70) System.out.println("Grade: C");
        else if (percentage >= 60) System.out.println("Grade: D");
        else if (percentage >= 40) System.out.println("Grade: E");
        else System.out.println("Grade: F - Fail");
    }

    void dayName(Scanner sc) {
        System.out.print("Enter day number (1-7): ");
```

```
    int day = sc.nextInt();
    switch (day) {
        case 1: System.out.println("Monday"); break;
        case 2: System.out.println("Tuesday"); break;
        case 3: System.out.println("Wednesday"); break;
        case 4: System.out.println("Thursday"); break;
        case 5: System.out.println("Friday"); break;
        case 6: System.out.println("Saturday"); break;
        case 7: System.out.println("Sunday"); break;
        default: System.out.println("Invalid Day");
    }
}

void multiplicationTable(Scanner sc) {
    System.out.print("Enter a number: ");
    int n = sc.nextInt();
    for (int i = 1; i <= 10; i++)
        System.out.println(n + " x " + i + " = " + (n * i));
}

void printNumbers(Scanner sc) {
    System.out.print("Enter N: ");
    int n = sc.nextInt();
    System.out.println("Numbers from 1 to " + n + ":");
    for (int i = 1; i <= n; i++) System.out.print(i + " ");
    System.out.println();
}

void naturalSum(Scanner sc) {
    System.out.print("Enter N: ");
    int n = sc.nextInt();
    int sum = 0;
    for (int i = 1; i <= n; i++) sum = sum + i;
    System.out.println("Sum = " + sum);
}

void fibonacci(Scanner sc) {
    System.out.print("Enter number of terms: ");
    int n = sc.nextInt();
    int a = 0, b = 1;
    System.out.println("Fibonacci Series:");
    for (int i = 1; i <= n; i++) {
        System.out.print(a + " ");
        int c = a + b;
        a = b;
        b = c;
    }
    System.out.println();
}

void factorial(Scanner sc) {
    System.out.print("Enter a number: ");
    int n = sc.nextInt();
    int fact = 1;
    for (int i = 1; i <= n; i++) fact = fact * i;
    System.out.println("Factorial = " + fact);
}
```

```
    void prime(Scanner sc) {
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        boolean isPrime = true;
        if (n <= 1) isPrime = false;
        else {
            for (int i = 2; i < n; i++) {
                if (n % i == 0) {
                    isPrime = false;
                    break;
                }
            }
        }
        if (isPrime) System.out.println(n + " is Prime");
        else System.out.println(n + " is Not Prime");
    }

    void maximum(Scanner sc) {
        System.out.print("Enter two numbers: ");
        int a = sc.nextInt(), b = sc.nextInt();
        if (a > b) System.out.println("Maximum = " + a);
        else System.out.println("Maximum = " + b);
    }

    void circleArea(Scanner sc) {
        System.out.print("Enter radius: ");
        double r = sc.nextDouble();
        double area = Math.PI * r * r;
        System.out.println("Area of Circle = " + area);
    }
}

public class StudentUtilitySystem {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        StudentUtility obj = new StudentUtility();
        int choice;

        do {
            System.out.println("\n===== STUDENT UTILITY SYSTEM =====");
            System.out.println("1. Student Details");
            System.out.println("2. Even or Odd");
            System.out.println("3. Largest of Three Numbers");
            System.out.println("4. Grade Calculator");
            System.out.println("5. Day Name");
            System.out.println("6. Multiplication Table");
            System.out.println("7. Print Numbers");
            System.out.println("8. Sum of Natural Numbers");
            System.out.println("9. Fibonacci Series");
            System.out.println("10. Factorial");
            System.out.println("11. Prime Number");
            System.out.println("12. Maximum of Two Numbers");
            System.out.println("13. Area of Circle");
            System.out.println("14. Exit");
            System.out.print("Enter your choice: ");
            choice = sc.nextInt();
```

```
            sc.nextLine();

            switch (choice) {
                case 1: obj.studentDetails(sc); break;
                case 2: obj.evenOdd(sc); break;
                case 3: obj.largest(sc); break;
                case 4: obj.grade(sc); break;
                case 5: obj.dayName(sc); break;
                case 6: obj.multiplicationTable(sc); break;
                case 7: obj.printNumbers(sc); break;
                case 8: obj.naturalSum(sc); break;
                case 9: obj.fibonacci(sc); break;
                case 10: obj.factorial(sc); break;
                case 11: obj.prime(sc); break;
                case 12: obj.maximum(sc); break;
                case 13: obj.circleArea(sc); break;
                case 14:
                    System.out.println("Thank you for using Student Utility System!");
                    break;
                default: System.out.println("Invalid Choice!");
            }
        } while (choice != 14);
        sc.close();
    }
}
```

7. Sample Output

Illustrative console output is shown below with white text on a black background.

```
===== STUDENT UTILITY SYSTEM =====
1. Student Details
2. Even or Odd
3. Largest of Three Numbers
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12. Maximum of Two Numbers
13. Area of Circle
14. Exit
Enter your choice: 2
Enter a number: 8
8 is Even

===== STUDENT UTILITY SYSTEM =====
Enter your choice: 8
Enter N: 5
Sum = 15

===== STUDENT UTILITY SYSTEM =====
Enter your choice: 14
Thank you for using Student Utility System!
```

---

8. Result

The Student Utility System provides 13 utility operations and an Exit option through a menu-driven Java program. input, performs the selected calculation or check, displays the result, and returns to the menu until Exit is selected. cases illustrate the expected output for representative operations.

Note: The program expects valid numeric input. Very large factorials or sums may exceed the range of the int data

The Student Utility System provides 13 utility operations and an Exit option through a menu-driven Java program. It reads the user's input, performs the selected calculation or check, displays the result, and returns to the menu until Exit is selected.